Mahdi Saeedi Nikoo

SOFTWARE ENGINEER

Software Engineer with several years of experience building scalable platforms, automation systems, and AI-powered applications. PhD in Software Engineering with a focus on AI-driven systems and large-scale model analytics. Experienced in designing modular architectures, developing data-intensive pipelines, and building end-to-end applications across back-end and frontend. Strong track record of delivering practical, high-impact solutions that improve efficiency, scalability, and user experience.

Professional Experience

JUN 2025 – PRESENT (PT)

Datory, Eindhoven, NL.
Software Engineer (freelance)

- **AI-based automation platform**
  Developing an AI-driven platform using AI Agents and RAG to automate interview workflows, including question generation, candidate interaction, and data analysis.

  Technologies: Python, JavaScript, Streamlit, React, FastAPI, SQL, Power BI, RAG, LLM

NOV 2025 – FEB 2026 (PT)

Independent Project
Software Engineer

- **Protein Function Prediction (CAFA6)**
  Developing ML-based models for protein classification as part of an international competition (CAFA6).

  Technologies: Python, ML, Bioinformatics, Data Analysis

AUG 2019 – NOV 2025 (FT)

Eindhoven University of Technology, Eindhoven, NL.
Software Engineer (PhD)

- **Toolchin within Arrowhead Tools project**
  Led the development of a toolchain within the Eclipse Arrowhead IIoT framework, comprising subsystems for model analytics, a centralized model repository, and an interactive visualization dashboard. The toolchain enabled engineers to search, inspect, and manage large process model repositories more efficiently.
- **Clone detector for process models**
  Developed a clone detection tool for business process models by extending the SAMOS model analytics platform, enabling reuse and refactoring support for BPMN modelers.
- **Analysis of process models in open-source**
  Built a scalable data pipeline to ingest, process, and analyze thousands of open-source business process models from GitHub, enabling large-scale model search, comparison, and reuse.
- **Recommenders for process model design**
  Designed and developed AI-powered recommendation engine (LLM-based) for BPMN modeling, providing design suggestions and reducing modeling time for modelers.
- **Process model repository system**
  Led the design and implementation of a repository system for retrieving process models from open-source sources, enabling model transformation, filtering, and advanced search capabilities.
- **Domain-specific modeling for Arrowhead Framework**
  Led the architecture and implementation of a domain-specific modeling tool with automated code generation, enabling engineers to generate production-ready components from high-level models.

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Technical Skills

Programming Languages
Python, Java, C#, C/C++, VB.Net, VBA, R, SQL

Modeling & Design
UML, SysML, BPMN, OOP, Design Patterns

Web & APIs
HTML, CSS, JavaScript, React, FastAPI, Streamlit, REST, JSON, XML

Database Management
PostgreSQL, MS SQL, ElasticSearch

Version Control
Git, GitHub

Build Tools
Maven, Gradle

Parsing & DSL Tools
Xtext, ANTLR

Embedded Systems
Arduino

Containerization & Cloud
Docker, Microsoft Azure, AWS

AI/ML & NLP
RAG, Transformers, scikit-learn, pandas, NumPy

Tools & Platforms
Linux, Windows, IDEs (VS Code, Eclipse, IntelliJ), MS Office

Personal Skills

- Strong analytical ability
- Excellent written and verbal communication
- Effective teamwork and collaboration
- Innovative problem-solving
- High adaptability and fast learning
- Strong critical thinking
- Interdisciplinary mindset
- Experience supervising and mentoring juniors

- **Process model analytics dashboard**
  Led design and development of a web-based analytics dashboard delivering interactive visualizations and actionable insights for process model analysis, to support data-driven decision-making.

  Technologies: Java, Python, R, BPMN, SysML, ML, IR, NLP, Git, Elasticsearch, Kibana, Docker, GitHub, LLM

APR 2014 – JUL 2019 (FT)

Spark Measurement Technologies - Calibration Lab, Ankara, Turkey
Software Engineer

- **SparkS, a scripting language for electronic test equipment, Apr 2014 – Mar 2017**
  Developed SparkS, a DSL and interpreter for automating calibration, with exception handling, modularity, and variability features to enhance script safety and reuse.
- **Metrology Blocks™, Sep 2017 – Aug 2019**
  Led development of a visual/textual web editor for the SparkS language within the Metrology.NET™ platform, providing block-based and code-based editing to reduce onboarding time and boost adoption. Metrology Blocks™ was later showcased as an innovator at the Measurement Science Conference (2018).
- **Digital Twin development for electronic test equipment, Oct 2018 – Jul 2019**
  Led development of data-driven digital-twin models that learned instrument command and response behavior, enabling virtual testing and reducing hardware dependency and calibration costs.
- **Automation tools, Apr 2014 – Jul 2019**
  Built several automation, reporting, and certification tools that streamlined data collection, reduced manual effort, and improved scalability and workflow efficiency for calibration engineers.

  Technologies: Google Blockly, CodeMirror, JavaScript, CSS, HTML, SCPI, L* algorithm, C#, .NET, ANTLR, UML, Git

SEP 2017 – FEB 2018 (FT)

Cal Lab Solutions, Inc., Denver, USA
Software Developer

- **Interactive editing environment for a scripting language - Sep 2017 – Feb 2018**
  Developed a visual/textual editor for a metrology language with seamless block–code translation, improving usability for both novice and expert users.
- **Full-Stack editor for laboratory accreditation scopes**
  Developed both client-side and server-side features for an open-source system that supports editing and maintaining accreditation scope documents used by calibration labs.

  Technologies: C#, .NET, MVVM architecture, Git, Google Blockly, CodeMirror, JavaScript, CSS, HTML

Honors & Awards

ACM Asian Regional Programming Contest (ACM-ICPC) 2008
Regional competitor – Tehran, Iran

National Collegiate Scientific Competition 2009
Selected competitor – Shiraz, Iran

TeknoTez Industry-Sponsored M.Sc. Thesis Grant 2014
METU Technopolis, Ankara, Turkey

Full Scholarship for Ph.D. Studies 2015
Middle East Technical University, Ankara, Turkey

Education

Ph.D. in Software Engineering 2019 – 2025
Eindhoven University of Technology, The Netherlands

M.Sc. in Software Engineering 2012 – 2015
Middle East Technical University, Turkey

B.Sc. in Software Engineering 2005 – 2009
University of Tabriz, Iran

Extracurricular Activities

- Coordinated social activities as a member of the PhD Council, Department of M&CS, TU/e (2022–2023).
- Facilitated and mentored software development for multiple internships, Bachelor's, and Master's projects.
- participated in multiple training sessions and workshops on software analytics, complexity, architecture design, and modeling.

Languages

ENGLISH	fluent
DUTCH	B1 - improving
AZERBAIJANI	native
PERSIAN	native
TURKISH	fluent

Personal interests

pencil drawing, jogging, cooking, music playing, table tennis